SG No. 140 D_{4h}^{18} $I4/mcm$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `4a`, site symmetry: `422`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `422`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,-5,-6,-7,-8]`
2	$[0,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12,-13,-14,-15,-16]`

Table 2: Wyckoff bond: `8b@4a`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,-2,-5,6]`
2	$[0,\ X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,-4,7,-8]`
3	$[-X,\ 0,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,-10,-13,14]`
4	$[0,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[11,-12,15,-16]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X,\ X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,-2,7,-8]`
2	$[-X,\ X,\ 0]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,-4,5,-6]`
3	$[-X,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,-10,15,-16]`
4	$[X,\ -X,\ 0]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[11,-12,13,-14]`

Table 4: Wyckoff bond: `16d@4a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,-5]`
2	$[-X,\ 0,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2,-6]`
3	$[0,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,-8]`
4	$[0,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[4,-7]`
5	$[-X,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,-13]`
6	$[X,\ 0,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[10,-14]`
7	$[0,\ -X,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[11,-16]`
8	$[0,\ X,\ -Z]$	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[12,-15]`

Table 5: Wyckoff bond: `16e@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,-8]`
2	$[-X,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[2,-7]`
3	$[-X,\ X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[3,-6]`
4	$[X,\ -X,\ Z]$	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[4,-5]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[9,-16]`
6	$[X, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[10,-15]`
7	$[X, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[11,-14]`
8	$[-X, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[12,-13]`

Table 6: Wyckoff bond: `16f@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, \frac{1}{4}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, 0, \frac{1}{4}]$	`[3,-4]`
3	$[-X, Y, 0]$	$[0, 0, \frac{1}{4}]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{4}]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[0, 0, \frac{3}{4}]$	`[9,-10]`
6	$[Y, -X, 0]$	$[0, 0, \frac{3}{4}]$	`[11,-12]`
7	$[X, -Y, 0]$	$[0, 0, \frac{3}{4}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[0, 0, \frac{3}{4}]$	`[15,-16]`

Table 7: Wyckoff bond: `32g@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, \frac{1}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, \frac{1}{4}]$	`[2]`
3	$[-Y, X, Z]$	$[0, 0, \frac{1}{4}]$	`[3]`
4	$[Y, -X, Z]$	$[0, 0, \frac{1}{4}]$	`[4]`
5	$[-X, Y, -Z]$	$[0, 0, \frac{1}{4}]$	`[5]`
6	$[X, -Y, -Z]$	$[0, 0, \frac{1}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{4}]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{4}]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, 0, \frac{3}{4}]$	`[9]`
10	$[X, Y, -Z]$	$[0, 0, \frac{3}{4}]$	`[10]`
11	$[Y, -X, -Z]$	$[0, 0, \frac{3}{4}]$	`[11]`
12	$[-Y, X, -Z]$	$[0, 0, \frac{3}{4}]$	`[12]`
13	$[X, -Y, Z]$	$[0, 0, \frac{3}{4}]$	`[13]`
14	$[-X, Y, Z]$	$[0, 0, \frac{3}{4}]$	`[14]`
15	$[-Y, -X, Z]$	$[0, 0, \frac{3}{4}]$	`[15]`
16	$[Y, X, Z]$	$[0, 0, \frac{3}{4}]$	`[16]`

* Wyckoff site: `4b`, site symmetry: `-42m`

Table 8: Wyckoff bond: `4a@4b`, bond symmetry: `-42m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[1,2,-5,-6,-11,-12,15,16]`
2	$[0, 0, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[3,4,-7,-8,-9,-10,13,14]`

Table 9: Wyckoff bond: `8b@4b`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,-4,7,-8]`
3	$[-X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-10,-13,14]`
4	$[0, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[11,-12,15,-16]`

Table 10: Wyckoff bond: `8c@4b`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2,-15,16]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,-4,-13,14]`
3	$[-X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[5,-6,-11,12]`
4	$[X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7,-8,-9,10]`

Table 11: Wyckoff bond: `16d@4b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[4,-7]`
5	$[-X, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-13]`
6	$[X, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[10,-14]`
7	$[0, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[11,-16]`
8	$[0, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[12,-15]`

Table 12: Wyckoff bond: `16e@4b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[4,13]`
5	$[-X, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[5,12]`
6	$[X, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[6,11]`
7	$[X, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7,10]`
8	$[-X, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[8,9]`

Table 13: Wyckoff bond: `16f@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,-4]`

continued ...

Table 13

No.	vector	center	mapping
3	$[-X, Y, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[5,-6]`
4	$[Y, X, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[9,-10]`
6	$[Y, -X, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[11,-12]`
7	$[X, -Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[15,-16]`

Table 14: Wyckoff bond: `32g@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[2]`
3	$[-Y, X, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[3]`
4	$[Y, -X, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[4]`
5	$[-X, Y, -Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[5]`
6	$[X, -Y, -Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[7]`
8	$[-Y, -X, -Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[8]`
9	$[-X, -Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[9]`
10	$[X, Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[10]`
11	$[Y, -X, -Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[11]`
12	$[-Y, X, -Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[12]`
13	$[X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[13]`
14	$[-X, Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[14]`
15	$[-Y, -X, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[15]`
16	$[Y, X, Z]$	$[0, \frac{1}{2}, \frac{1}{4}]$	`[16]`

* Wyckoff site: `4c`, site symmetry: `4/m..`

Table 15: Wyckoff bond: `4a@4c`, bond symmetry: `4/m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, 0]$	`[1,2,3,4,-9,-10,-11,-12]`
2	$[0, 0, -Z]$	$[0, 0, \frac{1}{2}]$	`[5,6,7,8,-13,-14,-15,-16]`

Table 16: Wyckoff bond: `8b@4c`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, 0]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$[0, 0, 0]$	`[3,-4,-11,12]`
3	$[-X, Y, 0]$	$[0, 0, \frac{1}{2}]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2}]$	`[7,-8,-15,16]`

Table 17: Wyckoff bond: `16c@4c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2,-10]`
3	$[-Y, X, Z]$	$[0, 0, 0]$	`[3,-11]`
4	$[Y, -X, Z]$	$[0, 0, 0]$	`[4,-12]`
5	$[-X, Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[5,-13]`
6	$[X, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[6,-14]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2}]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2}]$	`[8,-16]`

* Wyckoff site: `4d`, site symmetry: `m.mm`

Table 18: Wyckoff bond: `4a@4d`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, \frac{1}{2}, 0]$	`[1,-2,7,-8,-9,10,-15,16]`
2	$[-X, X, 0]$	$[\frac{1}{2}, 0, 0]$	`[3,-4,5,-6,-11,12,-13,14]`

Table 19: Wyckoff bond: `4b@4d`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[0, \frac{1}{2}, 0]$	`[1,-2,-7,8,-9,10,15,-16]`
2	$[X, X, 0]$	$[\frac{1}{2}, 0, 0]$	`[3,-4,-5,6,-11,12,13,-14]`

Table 20: Wyckoff bond: `4c@4d`, bond symmetry: `m.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, \frac{1}{2}, 0]$	`[1,2,-7,-8,-9,-10,15,16]`
2	$[0, 0, Z]$	$[\frac{1}{2}, 0, 0]$	`[3,4,-5,-6,-11,-12,13,14]`

Table 21: Wyckoff bond: `8d@4d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, \frac{1}{2}, 0]$	`[1,-8,-9,16]`
2	$[-X, -X, Z]$	$[0, \frac{1}{2}, 0]$	`[2,-7,-10,15]`
3	$[-X, X, Z]$	$[\frac{1}{2}, 0, 0]$	`[3,-6,-11,14]`
4	$[X, -X, Z]$	$[\frac{1}{2}, 0, 0]$	`[4,-5,-12,13]`

Table 22: Wyckoff bond: `8e@4d`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[0, \frac{1}{2}, 0]$	`[1,-7,-9,15]`
2	$[-X, X, Z]$	$[0, \frac{1}{2}, 0]$	`[2,-8,-10,16]`

continued ...

Table 22

No.	vector	center	mapping
3	$[X, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-5,-11,13]`
4	$[-X, -X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-6,-12,14]`

Table 23: Wyckoff bond: `8f@4d`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-2,-9,10]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-4,-11,12]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-6,-13,14]`
4	$[Y, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[7,-8,-15,16]`

Table 24: Wyckoff bond: `16g@4d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[8,-16]`

* Wyckoff site: `8e`, site symmetry: `..2/m`

Table 25: Wyckoff bond: `8a@8e`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,-7,-9,15]`
2	$[-X, X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,-8,-10,16]`
3	$[X, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[3,-5,-11,13]`
4	$[-X, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,-6,-12,14]`

Table 26: Wyckoff bond: `8b@8e`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,7,-9,-15]`
2	$[-X, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,8,-10,-16]`
3	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[3,5,-11,-13]`
4	$[X, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,6,-12,-14]`

Table 27: Wyckoff bond: `16c@8e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[2,-10]`
3	$[-Y, X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[3,-11]`
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[4,-12]`
5	$[-X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,-13]`
6	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6,-14]`
7	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[8,-16]`

* Wyckoff site: `8f`, site symmetry: `4..`

Table 28: Wyckoff bond: `8a@8f`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[0, 0, z]$	`[1,2,3,4]`
2	$[0, 0, -Z]$	$[0, 0, \frac{1}{2} - z]$	`[5,6,7,8]`
3	$[0, 0, -Z]$	$[0, 0, -z]$	`[9,10,11,12]`
4	$[0, 0, Z]$	$[0, 0, z + \frac{1}{2}]$	`[13,14,15,16]`

Table 29: Wyckoff bond: `16b@8f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, 0, z]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, 0, z]$	`[3,-4]`
3	$[-X, Y, 0]$	$[0, 0, \frac{1}{2} - z]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, 0, \frac{1}{2} - z]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[0, 0, -z]$	`[9,-10]`
6	$[Y, -X, 0]$	$[0, 0, -z]$	`[11,-12]`
7	$[X, -Y, 0]$	$[0, 0, z + \frac{1}{2}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[0, 0, z + \frac{1}{2}]$	`[15,-16]`

Table 30: Wyckoff bond: `32c@8f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, z]$	`[1]`
2	$[-X, -Y, Z]$	$[0, 0, z]$	`[2]`
3	$[-Y, X, Z]$	$[0, 0, z]$	`[3]`
4	$[Y, -X, Z]$	$[0, 0, z]$	`[4]`
5	$[-X, Y, -Z]$	$[0, 0, \frac{1}{2} - z]$	`[5]`
6	$[X, -Y, -Z]$	$[0, 0, \frac{1}{2} - z]$	`[6]`
7	$[Y, X, -Z]$	$[0, 0, \frac{1}{2} - z]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, 0, \frac{1}{2} - z]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, 0, -z]$	`[9]`
10	$[X, Y, -Z]$	$[0, 0, -z]$	`[10]`

continued ...

Table 30

No.	vector	center	mapping
11	$[Y, -X, -Z]$	$[0, 0, -z]$	`[11]`
12	$[-Y, X, -Z]$	$[0, 0, -z]$	`[12]`
13	$[X, -Y, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, 0, z+\frac{1}{2}\right]$	`[16]`

* Wyckoff site: `8g`, site symmetry: `2.mm`

Table 31: Wyckoff bond: `8a@8g`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2,-15,16]`
2	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,-4,-13,14]`
3	$[-X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[5,-6,-11,12]`
4	$[X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[7,-8,-9,10]`

Table 32: Wyckoff bond: `8b@8g`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,2,15,16]`
2	$[0, 0, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,4,13,14]`
3	$[0, 0, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[7,8,9,10]`

Table 33: Wyckoff bond: `16c@8g`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	`[4,13]`
5	$[-X, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[5,12]`
6	$[X, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[6,11]`
7	$[X, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[7,10]`
8	$[-X, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[8,9]`

Table 34: Wyckoff bond: `16d@8g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, z\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[5,-6]`

continued ...

Table 34

No.	vector	center	mapping
4	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[7,-8]
5	$[-X, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[9,-10]
6	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[11,-12]
7	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, z\right]$	[13,-14]
8	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, z\right]$	[15,-16]

Table 35: Wyckoff bond: 32e@8g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	[1]
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, z\right]$	[2]
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[3]
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[4]
5	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[6]
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[9]
10	$[X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[10]
11	$[Y, -X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[11]
12	$[-Y, X, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[12]
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[13]
14	$[-X, Y, Z]$	$\left[\frac{1}{2}, 0, z\right]$	[14]
15	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[15]
16	$[Y, X, Z]$	$\left[0, \frac{1}{2}, z\right]$	[16]

* Wyckoff site: 8h, site symmetry: m.2m

Table 36: Wyckoff bond: 8a@8h, bond symmetry: m.2m

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x + \frac{1}{2}, 0\right]$	[1,7,10,16]
2	$[-X, -X, 0]$	$\left[-x, \frac{1}{2} - x, 0\right]$	[2,8,9,15]
3	$[-X, X, 0]$	$\left[\frac{1}{2} - x, x, 0\right]$	[3,5,12,14]
4	$[X, -X, 0]$	$\left[x + \frac{1}{2}, -x, 0\right]$	[4,6,11,13]

Table 37: Wyckoff bond: 8b@8h, bond symmetry: m.2m

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[x, x + \frac{1}{2}, 0\right]$	[1,-7,10,-16]
2	$[-X, X, 0]$	$\left[-x, \frac{1}{2} - x, 0\right]$	[2,-8,9,-15]
3	$[X, X, 0]$	$\left[\frac{1}{2} - x, x, 0\right]$	[3,-5,12,-14]
4	$[-X, -X, 0]$	$\left[x + \frac{1}{2}, -x, 0\right]$	[4,-6,11,-13]

Table 38: Wyckoff bond: `8c@8h`, bond symmetry: `m.2m`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,-7,-10,16]`
2	$[0,\ 0,\ Z]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,-8,-9,15]`
3	$[0,\ 0,\ Z]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[3,-5,-12,14]`
4	$[0,\ 0,\ Z]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,-6,-11,13]`

Table 39: Wyckoff bond: `16d@8h`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,16]`
2	$[-X,\ -X,\ Z]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,15]`
3	$[-X,\ X,\ Z]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[3,14]`
4	$[X,\ -X,\ Z]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,13]`
5	$[-X,\ X,\ -Z]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[5,12]`
6	$[X,\ -X,\ -Z]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[6,11]`
7	$[X,\ X,\ -Z]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[7,10]`
8	$[-X,\ -X,\ -Z]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[8,9]`

Table 40: Wyckoff bond: `16e@8h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ -X,\ Z]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,-7]`
2	$[-X,\ X,\ Z]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,-8]`
3	$[X,\ X,\ Z]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[3,-5]`
4	$[-X,\ -X,\ Z]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,-6]`
5	$[-X,\ X,\ -Z]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[9,-15]`
6	$[X,\ -X,\ -Z]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[10,-16]`
7	$[-X,\ -X,\ -Z]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[11,-13]`
8	$[X,\ X,\ -Z]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[12,-14]`

Table 41: Wyckoff bond: `16f@8h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[1,10]`
2	$[-X,\ -Y,\ 0]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[2,9]`
3	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[3,12]`
4	$[Y,\ -X,\ 0]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[4,11]`
5	$[-X,\ Y,\ 0]$	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	`[5,14]`
6	$[X,\ -Y,\ 0]$	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	`[6,13]`
7	$[Y,\ X,\ 0]$	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[7,16]`
8	$[-Y,\ -X,\ 0]$	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[8,15]`

Table 42: Wyckoff bond: `32g@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x, 0\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x+\frac{1}{2}, -x, 0\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, 0\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, -x, 0\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, 0\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}-x, x, 0\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, 0\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x, 0\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, 0\right]$	`[15]`
16	$[Y, X, Z]$	$\left[x, x+\frac{1}{2}, 0\right]$	`[16]`

* Wyckoff site: `16i`, site symmetry: `..2`

Table 43: Wyckoff bond: `16a@16i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[x, x, \frac{1}{4}\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[4,-6]`
5	$[-X, X, -Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[9,-15]`
6	$[X, -X, -Z]$	$\left[x, x, \frac{3}{4}\right]$	`[10,-16]`
7	$[-X, -X, -Z]$	$\left[x, -x, \frac{3}{4}\right]$	`[11,-13]`
8	$[X, X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[12,-14]`

Table 44: Wyckoff bond: `16b@16i`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[x, x, \frac{1}{4}\right]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[-x, -x, \frac{1}{4}\right]$	`[2,8]`
3	$[-X, X, 0]$	$\left[-x, x, \frac{1}{4}\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x, -x, \frac{1}{4}\right]$	`[4,6]`
5	$[-X, -X, 0]$	$\left[-x, -x, \frac{3}{4}\right]$	`[9,15]`
6	$[X, X, 0]$	$\left[x, x, \frac{3}{4}\right]$	`[10,16]`
7	$[X, -X, 0]$	$\left[x, -x, \frac{3}{4}\right]$	`[11,13]`
8	$[-X, X, 0]$	$\left[-x, x, \frac{3}{4}\right]$	`[12,14]`

Table 45: Wyckoff bond: `32c@16i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, -x, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-x, -x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, x, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[x, -x, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, -x, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, x, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-x, -x, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[x, x, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `16j`, site symmetry: `.2.`

Table 46: Wyckoff bond: `16a@16j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[0, x, \frac{1}{4}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[4,-8]`
5	$[0, -X, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9,-14]`
6	$[0, X, -Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[10,-13]`
7	$[X, 0, -Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[11,-15]`
8	$[-X, 0, -Z]$	$\left[0, x, \frac{3}{4}\right]$	`[12,-16]`

Table 47: Wyckoff bond: `16b@16j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{4}\right]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{4}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[0, x, \frac{1}{4}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[0, -x, \frac{1}{4}\right]$	`[4,8]`
5	$[-X, 0, 0]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9,14]`
6	$[X, 0, 0]$	$\left[x, 0, \frac{3}{4}\right]$	`[10,13]`
7	$[0, -X, 0]$	$\left[0, -x, \frac{3}{4}\right]$	`[11,15]`
8	$[0, X, 0]$	$\left[0, x, \frac{3}{4}\right]$	`[12,16]`

Table 48: Wyckoff bond: `32c@16j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[0, x, \frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, 0, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, 0, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, x, \frac{1}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, -x, \frac{1}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, x, \frac{3}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, 0, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, -x, \frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, x, \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `16k`, site symmetry: `m..`

Table 49: Wyckoff bond: `16a@16k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[x, y, 0]$	`[1,10]`
2	$[-X, -Y, 0]$	$[-x, -y, 0]$	`[2,9]`
3	$[-Y, X, 0]$	$[-y, x, 0]$	`[3,12]`
4	$[Y, -X, 0]$	$[y, -x, 0]$	`[4,11]`
5	$[-X, Y, 0]$	$\left[-x, y, \frac{1}{2}\right]$	`[5,14]`
6	$[X, -Y, 0]$	$\left[x, -y, \frac{1}{2}\right]$	`[6,13]`
7	$[Y, X, 0]$	$\left[y, x, \frac{1}{2}\right]$	`[7,16]`
8	$[-Y, -X, 0]$	$\left[-y, -x, \frac{1}{2}\right]$	`[8,15]`

Table 50: Wyckoff bond: `16b@16k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[x, y, 0]$	`[1,-10]`
2	$[0, 0, Z]$	$[-x, -y, 0]$	`[2,-9]`
3	$[0, 0, Z]$	$[-y, x, 0]$	`[3,-12]`
4	$[0, 0, Z]$	$[y, -x, 0]$	`[4,-11]`
5	$[0, 0, -Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[5,-14]`
6	$[0, 0, -Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[6,-13]`
7	$[0, 0, -Z]$	$\left[y, x, \frac{1}{2}\right]$	`[7,-16]`
8	$[0, 0, -Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[8,-15]`

Table 51: Wyckoff bond: `32c@16k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, 0]$	`[2]`
3	$[-Y, X, Z]$	$[-y, x, 0]$	`[3]`
4	$[Y, -X, Z]$	$[y, -x, 0]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, 0]$	`[9]`
10	$[X, Y, -Z]$	$[x, y, 0]$	`[10]`
11	$[Y, -X, -Z]$	$[y, -x, 0]$	`[11]`
12	$[-Y, X, -Z]$	$[-y, x, 0]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, -y, \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, y, \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-y, -x, \frac{1}{2}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y, x, \frac{1}{2}\right]$	`[16]`

* Wyckoff site: `16l`, site symmetry: `..m`

Table 52: Wyckoff bond: `16a@16l`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[x, x+\frac{1}{2}, z\right]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[-x, \frac{1}{2}-x, z\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{2}-x, x, z\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[x+\frac{1}{2}, -x, z\right]$	`[4,13]`
5	$[-X, X, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[5,12]`
6	$[X, -X, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[6,11]`
7	$[X, X, -Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-z\right]$	`[7,10]`
8	$[-X, -X, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-z\right]$	`[8,9]`

Table 53: Wyckoff bond: `16b@16l`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[x, x+\frac{1}{2}, z\right]$	`[1,-16]`
2	$[-X, X, 0]$	$\left[-x, \frac{1}{2}-x, z\right]$	`[2,-15]`
3	$[X, X, 0]$	$\left[\frac{1}{2}-x, x, z\right]$	`[3,-14]`
4	$[-X, -X, 0]$	$\left[x+\frac{1}{2}, -x, z\right]$	`[4,-13]`
5	$[-X, -X, 0]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[5,-12]`
6	$[X, X, 0]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[6,-11]`
7	$[-X, X, 0]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-z\right]$	`[7,-10]`
8	$[X, -X, 0]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-z\right]$	`[8,-9]`

Table 54: Wyckoff bond: 32c@16l, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{2}, z\right]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-x, z\right]$	[2]
3	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x, z\right]$	[3]
4	$[Y, -X, Z]$	$\left[x+\frac{1}{2}, -x, z\right]$	[4]
5	$[-X, Y, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[6]
7	$[Y, X, -Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-z\right]$	[8]
9	$[-X, -Y, -Z]$	$\left[\frac{1}{2}-x, -x, \frac{1}{2}-z\right]$	[9]
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x, \frac{1}{2}-z\right]$	[10]
11	$[Y, -X, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[11]
12	$[-Y, X, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[12]
13	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, z\right]$	[13]
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x, z\right]$	[14]
15	$[-Y, -X, Z]$	$\left[-x, \frac{1}{2}-x, z\right]$	[15]
16	$[Y, X, Z]$	$\left[x, x+\frac{1}{2}, z\right]$	[16]

* Wyckoff site: 32m, site symmetry: 1

Table 55: Wyckoff bond: 32a@32m, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -y, z]$	[2]
3	$[-Y, X, Z]$	$[-y, x, z]$	[3]
4	$[Y, -X, Z]$	$[y, -x, z]$	[4]
5	$[-X, Y, -Z]$	$\left[-x, y, \frac{1}{2}-z\right]$	[5]
6	$[X, -Y, -Z]$	$\left[x, -y, \frac{1}{2}-z\right]$	[6]
7	$[Y, X, -Z]$	$\left[y, x, \frac{1}{2}-z\right]$	[7]
8	$[-Y, -X, -Z]$	$\left[-y, -x, \frac{1}{2}-z\right]$	[8]
9	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[9]
10	$[X, Y, -Z]$	$[x, y, -z]$	[10]
11	$[Y, -X, -Z]$	$[y, -x, -z]$	[11]
12	$[-Y, X, -Z]$	$[-y, x, -z]$	[12]
13	$[X, -Y, Z]$	$\left[x, -y, z+\frac{1}{2}\right]$	[13]
14	$[-X, Y, Z]$	$\left[-x, y, z+\frac{1}{2}\right]$	[14]
15	$[-Y, -X, Z]$	$\left[-y, -x, z+\frac{1}{2}\right]$	[15]
16	$[Y, X, Z]$	$\left[y, x, z+\frac{1}{2}\right]$	[16]